

Dhanush A

Senior Data Scientist | Generative AI & Data Engineering

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PROFESSIONAL SUMMARY:

Result driven Data Scientist 5+ years of experience in Generative AI, Machine Learning, and Data Engineering across banking, pharma, and retail. Skilled in GenAI workflows, RAG pipelines, RLHF optimization, and NLP solutions using Vertex AI, Hugging Face, and AWS/GCP. Proven expertise in ETL/ELT pipelines, real-time fraud detection, sentiment analysis, and executive dashboards, with strong focus on MLOps, CI/CD automation, and AI-driven business transformation.

CERTIFICATIONS:

IBM Data Science Certified

Microsoft AI

CERTIFIED DATA SCIENTIST

Microsoft Generate images with AI

TECHNICAL SKILLS:

Programming & Scripting	Python, SQL, Shell Scripting
Machine Learning / AI	Hugging Face Transformers, BERT, Random Forest, scikit-learn, A/B Testing, Model Fine-Tuning, Statistical Analysis
Generative AI & LLMs	LangChain, RAG (FAISS, ChromaDB), RLHF (PPO), Prompt Engineering, Semantic Search, OCR (AWS Textract)
Data Engineering / ETL	Spark, MySQL, ETL Development, Data Pipelines, Data Ingestion
MLOps & Automation	Vertex AI Pipelines, Docker, Kubernetes, Jenkins, GitLab CI, Artifact Registry, Git
Cloud Platforms	Google Cloud (Vertex AI, BigQuery, Dataflow, GKE, Cloud Functions), AWS (EC2, S3, Textract, Lambda, Glue, CloudWatch, Redshift), Azure (basic exposure)
Databases & Storage	BigQuery, MySQL, AWS Redshift, Cloud Storage Systems
Business Impact Areas	Customer Service Automation, Fraud Detection, KYC Automation, Pharma Analytics, Retail Insights

PROFESSIONAL EXPERIENCE:

First Citizens Bank, NC

May 2024 – Present

Role: Gen AI Data Scientist

Responsibilities:

- Developed scalable Generative AI workflows using LangChain and LangGraph for customer service automation and fraud detection.
- Implemented RAG pipelines using FAISS and ChromaDB for real-time, context-aware LLM responses from enterprise knowledge bases.
- Automated the ML lifecycle with CI/CD using Vertex AI Pipelines, enabling continuous retraining, deployment, and monitoring to adapt models to evolving fraud patterns and customer needs.
- Optimized RLHF training pipelines with PPO, reducing hallucinations and enhancing model response accuracy with custom relevance scoring.
- Orchestrated end-to-end MLOps workflows by Vertex AI, Cloud Functions, and Artifact Registry with monitoring & automated delivery.
- Built OCR-driven document intelligence solutions with AWS Textract and semantic search to streamline KYC & billing dataset processing.

GSk Pharmaceuticals, NC

Aug 2023 – May 2024

Role: Data Scientist

Responsibilities:

- Fine-tuned BERT models on Vertex AI for customer sentiment classification, improving feedback categorization and accelerating response prioritization.
- Built and deployed ML models on Google Kubernetes Engine (GKE), enabling real-time transaction monitoring and strengthening fraud risk mitigation.
- Engineered robust ETL pipelines with Python, SQL, and Cloud Dataflow, increasing data availability for analytics while minimizing manual intervention.
- Delivered executive-ready dashboards in Looker and Tableau, visualizing KPIs from BigQuery and clinical systems to support faster data-driven decision-making.

Flipkart, Bengaluru, India

Apr 2020 – July 2022

Role: Python Developer

Responsibilities:

- Fine-tuned ML models using Python and PyTorch/TensorFlow for retail-focused NLP applications, applying manual feature engineering and compliance validation to improve response accuracy and reliability.
- Developed data ingestion pipelines with Python, SQL, and Spark on Google Cloud Dataflow, enabling stakeholders to access product insights faster than with legacy systems.
- Streamlined data transformation and cleansing workflows using modular Python scripting and Google Cloud Functions, improving efficiency across ML and BI processes.
- Built a retail analytics platform on Google BigQuery with serverless integration, enabling interactive, low-latency data querying and reducing friction for analysts.
- Containerized NLP microservices with Docker and orchestrated deployment pipelines using Jenkins and GitLab CI, ensuring consistent environment replication across dev and production.
- Conducted A/B testing and statistical model evaluation with scikit-learn, strengthening reliability and robustness of models on dynamic retail datasets.
- Designed executive dashboards in Tableau connected to BigQuery and ETL outputs, accelerating reporting cycles and improving decision-making speed.

Fission Lab, Hyderabad, India

Mar 2019 – Apr 2020

Role: Software Developer

Responsibilities:

- Engineered ETL pipelines using Python and shell scripting, streamlining data migration across MySQL and cloud storage systems to improve processing efficiency.
- Developed and exposed REST APIs with Flask and Django REST Framework, automating data workflows and enhancing delivery speed and reliability.
- Deployed data services on AWS EC2 and integrated CloudWatch monitoring and alerts, minimizing downtime and improving system availability.
- Implemented batch ingestion pipelines for CSV, JSON, and XML formats using Amazon S3 for interim storage and backups, improving data throughput and reliability.
- Leveraged Git and CI/CD pipelines to accelerate deployment cycles, efficiently resolve UAT issues, and ensure on-time delivery for multiple client releases.

EDUCATION:

- Masters in Computer Science – Governor State University, University Park, IL, USA - 2024